

MUHAMMAD HISYAM

Full Stack & AI Software Engineer

Email: muhammad2504hisyam@gmail.com · GitHub: github.com/caleheinz25 · Portfolio: mhisyam.dev · Location: Indonesia

PROFESSIONAL SUMMARY

Full Stack & AI Software Engineer with strong experience in building high-performance web applications, real-time backend microservices, deep learning computer vision systems, and AI integration tools. Proficient in TypeScript, JavaScript, Python, Golang, Java, and PHP/Laravel. Creator of 26+ open-source and full-stack projects spanning medical AI segmentation, real-time IoT monitoring, Model Context Protocol (MCP) clients, and interactive web tools.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Golang, Java, PHP, SQL, HTML5, CSS3, LaTeX

Frontend Frameworks: Astro 5, React.js, SolidJS, SvelteKit, Next.js, Vue.js, Tailwind CSS

Backend & Cloud: Node.js, Express.js, Go (Gin/Fiber), Java Spring Boot, Laravel, REST APIs, WebSockets

AI & Data Engineering: PyTorch, U-Net, OpenCV, Model Context Protocol (MCP), Gemini API

Databases & DevOps: PostgreSQL, MongoDB, Redis, Docker, Git, GitHub Actions, Vite, Linux CLI

FEATURED PROJECTS

Developer Portfolio & Interactive Hub (Astro 5 + SolidJS)

[GitHub Repository](#)

- Engineered a high-speed developer portfolio using Astro 5, SolidJS, and Tailwind CSS 4.
- Integrated an embedded Dev Suite featuring a retro Unix terminal shell (`hisyam-cli`), REST API response demo client, audio music player, and 8 mini-games arcade.

Pneumonia Segmentation U-Net Model (Python + Deep Learning)

[GitHub Repository](#)

- Developed a deep learning semantic segmentation pipeline utilizing PyTorch and U-Net architecture for chest X-ray pneumonia detection.
- Implemented custom loss metrics, data augmentation transformers, and inference evaluation routines.

Gemini Model Context Protocol (MCP) Client (TypeScript)

[GitHub Repository](#)

- Built an extensible MCP client enabling agentic AI workflows and tool-use integrations with Google Gemini models.

Real-Time Energy Monitoring System (TypeScript + IoT)

[GitHub Repository](#)

- Built a real-time energy telemetry analytics dashboard processing high-frequency sensor streams with dynamic live charts.

Golang & SvelteKit Real-Time Chat App (Go + WebSockets)

[GitHub Repository](#)

- Implemented a concurrent real-time chat application with a Golang WebSocket backend server and SvelteKit reactive frontend.

Secure JWT Authentication with Refresh & Version Tokens (Java)

[GitHub Repository](#)

- Designed a robust Java authentication microservice supporting access tokens, sliding session refresh tokens, and instant token version invalidation.

EDUCATION & COMMUNITY

Bachelor of Science in Computer Science / Information Technology

Indonesia

Focus on Software Engineering, Web Systems, and Applied AI

- Maintainer of 26+ open-source GitHub repositories spanning AI, web apps, backend APIs, and developer utilities.
- Contributor to collaborative software projects including SMKN 32 SmartAssist AI Chatbot and Woven web platform.